

AGENTIC AI ARCHITECTURE

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Your Household Runs on AI Logic

Every concept in a multi-agent AI system has a perfect real-world counterpart — right in your home. Here's the full technical map.

 Household Team

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 Multi-Agent AI System

LAYER 01 — COORDINATION

 ORCHESTRATOR

The House Manager

You — the person who coordinates the whole household

The orchestrator is the top-level controller that receives a high-level goal, decomposes it into tasks, assigns those tasks to the right agents, monitors progress, handles conflicts, and assembles the final result. It doesn't do the work itself — it directs.

 **Planning**
Breaking a big goal into ordered sub-tasks, anticipating dependencies and sequencing steps before execution begins.

 You plan: clean before guests arrive, cook after groceries are delivered, driver picks up kids at 3pm.

 **Task Routing**
Dispatching the right sub-task to the right specialist agent based on capability, availability, and context.

 You don't ask the driver to cook. Each person gets their domain.

 **Conflict Resolution**
Detecting when two agents' plans would clash and re-sequencing or re-assigning to avoid it.

 Ensuring the cleaner finishes before the kids arrive home — not after.

 **Human-in-the-Loop**
Pausing execution to request human approval before taking high-stakes or irreversible actions. Keeps humans as the final authority.

 The driver calls you before taking an unfamiliar route during a storm.



AGENTS

The Specialist Workers

Driver, Cook, Cleaner — each owns a domain

Agents are autonomous AI entities with a defined goal, a set of tools, and the ability to reason and act in a loop until the task is complete. Each agent is specialized — it knows its domain deeply and operates independently.



AGENT

The Driver

Handles all mobility tasks — route planning, timing, vehicle selection. Gets you where you need to go, safely and efficiently.

Navigation Tool Use
Long-term Memory



AGENT

The Cook

Plans meals, sources ingredients, executes recipes. Adapts based on dietary preferences, what's in the fridge, and who's eating.

Planning Retrieval Adaptation



AGENT

The Cleaner

Maintains the home environment. Sequences rooms by priority, uses the right tools for each surface, and coordinates timing with other agents.

Sequencing Tool Use Timing



AGENT (EXTENDED)

The Shopper

Browses available options, compares prices, places orders, and tracks delivery status — an agent that interfaces with external systems.

Web Search API Calls
Decision Making



AGENT (EXTENDED)

The Health Monitor

Watches family health patterns, books appointments, manages prescriptions — long-running agent that tracks state over time.

State Tracking Scheduling
Episodic Memory



MEMORY SYSTEMS

How Agents Remember

The rules you told the driver — and the grocery list on the fridge

Memory is what separates a one-shot AI response from a true agent. Different memory types serve different purposes — some are fast and temporary, others are slow and permanent.



Long-term Memory

Persistent facts about preferences, rules, and learned patterns — stored across sessions and retrieved when relevant.



"Always take the highway after 8am." The driver never forgets this rule.



Working Memory

Short-term context held within a single task session — the agent's "active awareness" of what's happening right now.



The cook remembering what's already on the stove while chopping vegetables.



Episodic Memory

Records of past events and their outcomes — used to learn from experience and avoid repeating mistakes.



"Last time we ordered from that restaurant, delivery took 2 hours — avoid it on busy nights."



RAG — Retrieval-Augmented Generation

Agents dynamically look up information from external sources before generating a response — combining stored knowledge with real-time retrieval.



Cook checks a recipe book mid-task. Driver checks live traffic before departing.



TOOL USE

Choosing the Right Instrument

2-wheeler vs 4-wheeler — the driver picks the optimal vehicle for the job

Tool calling lets agents extend beyond their base capabilities — executing code, calling APIs, searching the web, reading files, sending messages. The agent decides which tool to use based on context, just like a skilled worker chooses the right instrument.



Web Search

Query live internet data — current prices, news, schedules, availability — that wasn't in the training data.



Driver checks live traffic and fuel prices before the trip.



Code Execution

Write and run code to solve precise computational tasks — calculations, data processing, file manipulation.



Cook calculates exact ingredient scaling for 12 guests instead of the recipe's 4.



API / External Services

Connect to calendars, payment systems, databases, IoT sensors, and third-party platforms to take real-world actions.



Shopper agent places an order via delivery app API, gets confirmation.



Adaptive Tool Selection

Agents evaluate context before picking a tool — choosing the most efficient path given current constraints.



Heavy rain + 4 passengers = 4-wheeler. Solo errand nearby = 2-wheeler.



FEEDBACK LOOPS

Checking & Improving

The cook tasting before serving — the driver re-routing mid-journey

Agentic systems don't just execute once — they loop, evaluate, and adjust. Reflection and feedback are what allow agents to catch errors, improve quality, and handle unexpected situations without human intervention.



Reflection / Self-Critique

An agent reviews its own output against the original goal before passing it on — catching errors proactively rather than reactively.



The cleaner checks each room against a mental checklist before calling it done.



Multi-Agent Communication

Agents share status, pass results, and negotiate handoffs — enabling collaborative workflows that no single agent could handle alone.



Cleaner tells cook "kitchen's done, you can start prepping" — not waiting for you to relay.



ReAct Loop (Reason + Act)

The core agentic cycle: Observe → Think → Act → Observe again. Agents iterate until the goal is achieved or a stop condition is met.



Driver observes traffic jam → thinks of alternate route → takes it → observes again.



Evaluation & Scoring

Automated quality checks compare outputs against criteria — used for self-improvement and to decide if a result is good enough to pass on.



Cook rates the dish before serving — "needs more salt" triggers a correction.

COMPLETE REFERENCE — ALL CONCEPTS MAPPED

AI CONCEPT	HOUSEHOLD ANALOGY	TECHNICAL DEFINITION	REAL-WORLD AI EXAMPLE
Orchestrator	You — the homeowner who coordinates everyone	Top-level controller that decomposes goals, routes tasks to agents, and assembles outputs	LangGraph StateGraph, AutoGen GroupChat manager, CrewAI Crew
Agent	Driver, Cook, Cleaner — each specialist	Autonomous entity with a goal, tools, and a reasoning loop (LLM + actions)	A coding agent, a research agent, a data analyst agent
Long-term Memory	Driver's permanent rules ("always highway after 8am")	Persistent vector store or database of facts that persists across sessions	Mem0, Zep, Pinecone with user profile embeddings
Working Memory	Cook's active awareness of what's currently on the stove	In-context window holding current task state — temporary, session-scoped	The LLM's context window during a task run
Episodic Memory	"That restaurant was late last time — avoid on busy nights"	Structured record of past events + outcomes, used to influence future decisions	Agent logging tool calls + results for future reference
RAG extended	Cook checks a recipe book; driver checks live traffic	Retrieve relevant external documents at inference time to ground generation in facts	LlamaIndex, LangChain retrieval chains, Perplexity AI
Tool Use / Calling	Choosing 2-wheeler vs 4-wheeler based on context	Agent selects and invokes structured functions (search, code, API) to extend its abilities	OpenAI function calling, Anthropic tool use, MCP servers
Planning extended	You plan: clean → cook → pick up kids, in order	Decomposing a high-level goal into ordered, dependency-aware sub-tasks before acting	ReAct, Tree of Thoughts, LLM-based task decomposition
Reflection extended	Cook tastes before serving; cleaner re-checks the room	Agent critiques its own output against the goal — iterating before passing results on	Reflexion framework, self-critique prompts in chain-of-thought
ReAct Loop extended	Driver sees traffic jam → re-routes → observes again	Interleaved Reason + Act cycle: Observe → Think → Act → Observe, until goal met	ReAct paper (Yao et al. 2022), standard in LangChain agents
Multi-Agent Comms extended	Cleaner tells cook "kitchen's ready, you can start"	Structured message-passing between agents to coordinate handoffs and share state	AutoGen conversations, CrewAI agent messaging, A2A protocol
Human-in-the-Loop extended	Driver calls you before taking an unfamiliar storm route	Pausing execution to request human approval on high-stakes or irreversible actions	LangGraph interrupt nodes, Slack approval bots, review queues
Guardrails extended	House rules: no strangers inside; don't open the gate at night	Hard constraints and safety filters that prevent agents from taking harmful or out-of-scope actions	NeMo Guardrails, constitutional AI rules, output validators